

Unit III – Conductors and Superconductors

Introduction

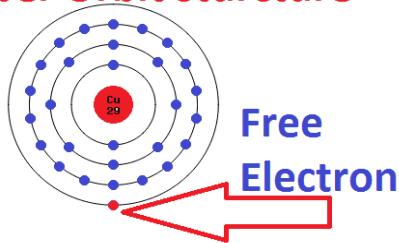
The free electron theory of metals aims to explain the structure and properties of solids through their electronic structure. The free electron theory is applicable to all solids i.e., both metals and non-metals. It explains the electrical and thermal properties of solids etc.

Concept of free electron:

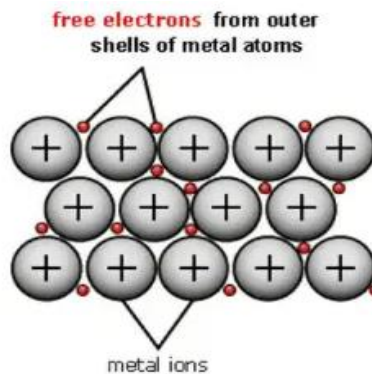
According to classical free electron theory, a metal can be considered to be made up of ion cores comprising of nucleus and inner electrons excluding valence electrons. These ion cores are immobile and do not take part in conduction process. The valence electrons are considered to be free and almost independent of ions to which they belong. These valence electrons can move under the influence of an applied electric field and hence contribute to the conductivity.

Thus, **free electrons are defined as the valence electrons which are responsible for the conduction in metals when they are influenced by the external electric field.**

Copper Orbit structure



k=2, l=8, m= 18, Last =1



Drift velocity (v_d)

If no electric field is applied on a conductor, the free electrons move in random directions. They collide with each other and also with the positive ions. Since the motion is completely random, average velocity in any direction is zero. If a constant electric field is applied inside a conductor, the electrons experience a force $F = -eE$ due to which they move in the direction opposite to direction of the field. As a result, in addition to the random motion, the electrons are subjected to a very slow directional motion. **This motion is called drift and the average velocity of this motion is called drift velocity v_d .**

$$v_d = \frac{eE\tau}{m}$$

where E is the applied electric field, e is the charge of an electron, m is its mass and τ is the collision time.

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Mobility of electrons (μ)

The mobility of electrons is defined as the drift velocity acquired per unit applied electric field.

$$\mu = \frac{v_d}{E} \quad \text{or} \quad \mu = \frac{e\tau}{m}$$

E is the applied electric field, v_d is the drift velocity. The unit of mobility is $\text{m}^2\text{V}^{-1}\text{s}^{-1}$

Mean free path (λ)

The average distance traveled by the electron between two successive collisions with lattice point is called mean free path.

$$\lambda = v \cdot \tau$$

v is the velocity of the electron, τ is the collision time.

Mean Collision time (τ)

It is defined as the average time taken by the free electrons between two successive collisions with the lattice point.

$$\tau = \frac{\lambda}{v}$$

λ is the mean free path and v is the velocity of the electrons.

Current density (J):

It is defined as the magnitude of current passing through unit area.

$$J = \frac{I}{A}$$

Where I is the current and A is the area of cross-section

Relation between J , σ and E :

From Ohm's law $V = IR$, we get the relation $\mathbf{J} = \sigma \mathbf{E}$

Thermal velocity (v_{th}):

The velocity with which the free electrons keep moving due to thermal agitation is called thermal velocity.

$$v_{th} = \sqrt{\frac{3k_B T}{m}}$$

k_B is the Boltzmann constant, m is the mass of the electron

Relaxation time (τ_r):

It is defined as the time taken by the free electrons to reach its equilibrium position from the disturbed position in the presence of electric field.

Expression for Relaxation time (τ_r):

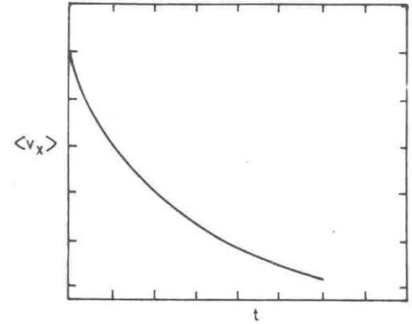
When the metal is subjected to an external electric field, the electrons move opposite to the applied field. After removal of electric field, the drift velocity decays exponentially as,

$$\langle v \rangle_t = \langle v \rangle_0 \exp(-t/\tau_r) \quad (1)$$

where $\langle v \rangle_0$ is the initial velocity when the field is switched off, $\langle v \rangle_t$ is the instantaneous velocity at any instant of time t and τ_r is a constant called 'Relaxation time'.

In Eq. (1), if $t = \tau_r$ then $\langle v \rangle_t = \frac{1}{e} \langle v \rangle_0$

Thus, the relaxation time may be stated as **the time taken for the drift velocity to decay to 1/e times of its original initial value.**



Classical free electron theory

Drude and Lorentz proposed this theory in 1900. According to this theory, the metals containing the free electrons obey the laws of classical mechanics

Assumptions of classical free electron theory:

1. Atoms in a metal are considered to be made up of ion cores which are nuclei surrounded by inner electrons excluding the valance electrons and the valance electrons are free to move anywhere inside the sample, which are responsible for electrical conductivity.
2. The free electrons are treated as equivalent to gas molecules, and they are assumed to obey the laws of kinetic theory of gases.

In the absence of field, as per kinetic theory of gases, the average energy with each electron at a temperature T will be $\frac{3}{2} k T$. It is related to kinetic energy through the relation

$$\frac{3}{2} kT = \frac{1}{2} m v_{th}^2$$

where v_{th} is the thermal velocity and k is the Boltzmann constant

3. The effect of electric potential due to positive ion cores on the electrons is considered to be constant and hence neglected.
4. The attraction between free electrons and the lattice ions, and the mutual repulsion among the electrons themselves is ignored, so that they move in all the directions with all possible velocities.

Expression for electrical conductivity in metals:

Consider a sample of conductor. The application of an electric field E will result in the current density J in a metal sample of conductivity σ given by Ohm's law,

$$J = \sigma E \quad \text{---- (1)}$$

As long as the applied electric field is held constant, the current density will be constant since the conductivity is a material constant.

The current density J , which is a measure of charge flowing through unit area of cross section in unit time, results from the motion of electrons and is given by

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$$J = -nev \quad \text{---- (2)}$$

where n is the number of free electrons in the metal per unit volume, e is the charge on the electron and v is the velocity acquired by the electrons in the electric field.

Assuming metal as equivalent to an electron gas, application of an electric field E produces an acceleration in the electron and may be written as equal to

$$a = -eE/m$$

Force acting on the electron $F = -eE = ma$, where m is the mass of the electron.

$$a = \left[\frac{\partial v}{\partial t} \right]_{Field} = \frac{-eE}{m}$$

Considering all the electrons in the metal, we may write

$$a = \left[\frac{\partial \langle v \rangle}{\partial t} \right]_{Field} = \frac{-eE}{m} \quad \text{----- (3)}$$

Where $\langle v \rangle$ is the average velocity of all the electrons taken together.

When the electric field E is held constant, the current density J will be constant from eqn (1).

$$\frac{dJ}{dt} = 0$$

$$\text{i.e., } \frac{d(-nev)}{dt} = 0 \quad \text{or} \quad \frac{d\langle v \rangle}{dt} = 0 \quad \text{----- (4)}$$

But from eqn (3), it is seen that the acceleration has a finite value which is not zero.

Hence, it is concluded that the electrons are acted upon by some other force also so that the resultant acceleration is zero.

$$\text{or } \frac{d\langle v \rangle}{dt} = \left[\frac{\partial \langle v \rangle}{\partial t} \right]_{Field} + \left[\frac{\partial \langle v \rangle}{\partial t} \right]_{El-Lattice} = 0 \quad \text{----- (5)}$$

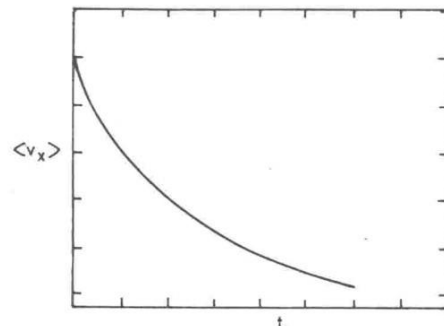
The second term is the deceleration term which must be arising due to the interaction of the electrons with the lattice, which can be calculated as follows:

Consider a metal sample to which an electric field E is applied. If the field is switched off at any instant of time, say $t = 0$, the current density also decreases to zero and the velocity also decreases to zero.

The variation of velocity with time may be represented by the equation,

$$\langle v \rangle_t = \langle v \rangle_0 \exp(-t/\tau) \quad \text{----- (6)}$$

where $\langle v \rangle_0$ is the initial velocity when the field is switched off, $\langle v \rangle_t$ is the instantaneous velocity at any



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instant of time t and τ is a constant called 'Relaxation time'.

Differentiating eqn. (6) with respect to time,

$$\left[\frac{\partial \langle v \rangle}{\partial t} \right]_{El-Lattice} = \frac{-\langle v \rangle_0}{\tau} e^{(-t/\tau)} = \frac{-\langle v \rangle_t}{\tau} \quad \text{----- (7)}$$

Substituting for the acceleration and deceleration terms from eqns. (3) and (7) in eqn. (5), we get

$$\frac{d\langle v \rangle}{dt} = \frac{-eE}{m} - \frac{-\langle v \rangle_t}{\tau} = 0$$

$$\text{Or } \langle v \rangle = \frac{-eE\tau}{m} \quad \text{----- (8)}$$

$$\text{Hence current density } J = -ne\langle v \rangle = -ne \left(\frac{-eE\tau}{m} \right) = \frac{ne^2E\tau}{m} \quad \text{----- (9)}$$

$$\text{From eqn. (1), we get, } J = \sigma E = \frac{ne^2E\tau}{m}$$

$$\sigma = \frac{ne^2\tau}{m}$$

This is the expression for the electrical conductivity of the metal.

Relation between conductivity and mobility:

$$\text{We have } \sigma = \frac{ne^2\tau}{m} = ne \left(\frac{e\tau}{m} \right) = ne\mu .$$

$$\text{Hence } \sigma = ne\mu$$

Failure of classical free electron theory

Electrical and thermal conductivities can be explained from classical free electron theory. But it fails to account the facts such as specific heat, temperature dependence of conductivity and dependence of electrical conductivity on electron concentration.

1. Temperature dependence of electrical conductivity:

Experimentally, electrical conductivity σ is inversely proportional to the temperature T .

$$\text{i.e. } \sigma_{exp} \propto \frac{1}{T} \quad \text{----- (1)}$$

but according to classical free electron theory, since,

$$\sigma = (ne^2\tau/m), \text{ and } \tau = \frac{\lambda}{v}, \text{ and } v = \sqrt{\frac{3kT}{m}}, \text{ it follows that,}$$

$$\sigma = \frac{ne^2\lambda}{\sqrt{3mkT}}$$

$$\text{and hence, } \sigma \propto \frac{1}{\sqrt{T}} \quad \text{----- (2)}$$

From equations (1) & (2) it is clear that the experimental value is not agreeing with the theory.

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2. Specific heat:

The molar specific heat of a gas at constant volume is, $C_v = \frac{3}{2}R$

As per the classical free electron theory, free electrons in a metal are expected to behave just as gas molecules. Thus, the above equation holds good equally well for the free electrons also.

But experimentally it was found that, the contribution to the specific heat of a metal by its conduction electrons was $C_v = 10^{-4} R$

which is far lower than the expected value.

3. Dependence of electrical conductivity on electron concentration:

According to the classical free electron theory,

$$\sigma = \frac{ne^2\tau}{m}$$

here n is the electron concentration, this indicates, $\sigma \propto n$

Consider two metals zinc (divalent element) and copper (monovalent element). Their electrical conductivities are $1.09 \times 10^7 / \Omega\text{m}$ and $5.88 \times 10^7 / \Omega\text{m}$ respectively. The electron concentrations for zinc is $13.1 \times 10^{28} / \text{m}^3$ which are much higher than that for copper $8.45 \times 10^{28} / \text{m}^3$. Hence the classical free electron theory fails to explain the dependence of electrical conductivity on electron concentration.

4. The mean free path of electrons in any metal calculated based on Drude's model was ten times less than experimentally observed value. This is because, Drude and Lorentz had assumed that the free electrons collide with the metal ions, not with the much farther apart lattice defects. The net result was resistivity values were 10 times greater than the measured values.

5. All good conductors are also good conductors of heat. This fact is given in the form of Wiedemann Franz law. This law states that "The ratio of thermal conductivity (κ) to electrical conductivity (σ) is proportional to the absolute temperature".i.e;

$$\frac{\kappa}{\sigma} \propto T.$$

But, experimentally it is found that Wiedemann Franz law is not true at low temperatures. At low temperature, the electrical conductivity (σ) and thermal conductivity (κ) vary in different ways. Therefore $\kappa/\sigma T$ is not a constant. But in classical free electron theory, it is a constant in all temperature.]

Quantum free electron theory:

The assumptions of quantum free electron theory are:

1. The energy values of the conduction electrons are quantized. The allowed energy values are realized in terms of a set of energy values.
2. The distribution of electrons in the various allowed energy levels occur as per Pauli's exclusion principle.
3. The electrons travel with a constant potential inside the metal but confined within its boundaries.

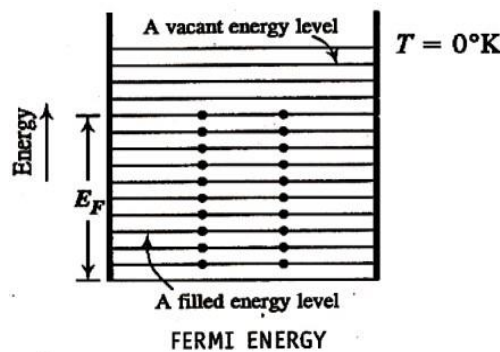
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4. The attraction between the electrons and the lattice ions and the repulsion between the electrons themselves are ignored.

Concept of Energy levels:

A metal which is having N number of atoms will have N allowed energy levels in each band. These energy levels are quantized and the distribution of electrons in the various allowed energy levels occur as per Pauli's exclusion principle. i.e. each energy level can accommodate a maximum of two electrons, one with spin up and the other with spin down. The filling up of energy levels occurs from the lowest level. The next pair of electrons occupies the next energy level and so on till all the electrons in the metal are accommodated.

Fermi energy & Fermi level: The energy of the highest occupied energy level at absolute zero temperature (0 K) is called the **Fermi energy** denoted by E_f and the corresponding energy level is called **Fermi level**.



There are three statistics

1. Maxwell- Boltzmann statistics:
 - Deals with particles which has no spin. Ex: Gaseous particles
2. Bose-Einstein statistics:
 - Deals with particles which has integral spin. Ex: Photons
3. Fermi-Dirac statistics:
 - Deals with particles which has half integral spin
 - Also known as Fermions
 - Ex: Electrons

Fermi factor: It is defined as the probability of occupation of a given energy state for a material in thermal equilibrium and is denoted by $f(E)$.

$$f(E) = \frac{1}{1 + e^{\left(\frac{E - E_f}{kT}\right)}}$$

where E_f is the fermi level k is the Boltzmann constant, T is the steady temperature.

Dependence of Fermi factor on temperature:

The dependence of Fermi factor on temperature and energy is as shown in the figure.

Following are the different cases:

1. Probability of occupation for $E < E_f$ at $T = 0$ K:

When $T = 0$ K and $E < E_f$

$$f(E) = \frac{1}{1 + e^{-\infty}} = \frac{1}{1 + 0} = 1$$

Thus, $f(E) = 1$ for $E < E_f$

This indicates at $T = 0$ K all the energy levels below the Fermi level are occupied.

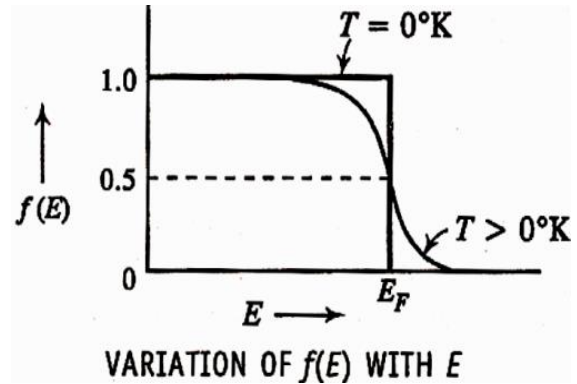
2. Probability of occupation for $E > E_f$ at $T = 0$ K:

When $T = 0$ K and $E > E_f$

$$f(E) = \frac{1}{1 + e^{\infty}} = \frac{1}{\infty} = 0$$

Thus, $f(E) = 0$ for $E > E_f$

This indicates that at $T = 0$ K, all the energy levels above Fermi levels are unoccupied.



3. The probability of occupation at ordinary temperature ($T > 0$ K): At ordinary temperatures, some of the electrons from energy levels below the Fermi level excite to the corresponding energy levels above the Fermi level. Thus $f(E)$ value decreases from 1 for the energy levels which are below the Fermi level, and the value of $f(E)$ increases from 0 for the energy levels which are above the Fermi level as shown in the figure.

At $E = E_f$ for $T > 0$ K

$$f(E) = \frac{1}{1 + e^0} = \frac{1}{1 + 1} = \frac{1}{2}$$

Thus, at ordinary temperature, the probability of occupation $f(E)$ is 0.5 or 50%

Fermi temperature (T_F):

Fermi temperature is defined as the temperature at which the average thermal energy of the free electrons equal to the fermi energy at 0 K,

$$T_f = \frac{E_f}{k_B}$$

where E_f is the fermi energy and k_B is the Boltzmann constant.

Fermi velocity (v_F):

The velocity of the electrons which occupy the fermi level is called fermi velocity.

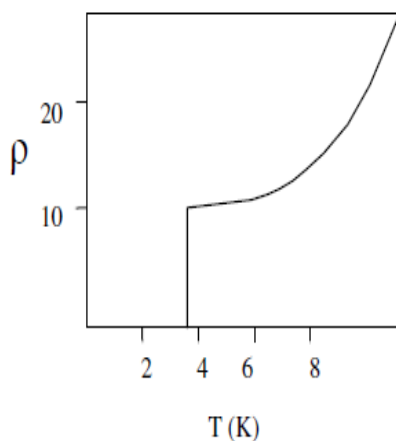
$$v_F = \left(\frac{2E_f}{m} \right)^{\frac{1}{2}}$$

where E_f is the fermi energy and m is the mass of electron.

Superconductors

Superconductivity

Certain metals and alloys exhibit almost zero resistivity (i.e. infinite conductivity), when they are cooled to sufficiently low temperatures. This effect is called superconductivity. The materials which exhibit this property are called superconductors. This phenomenon was first discovered by H. K. Onnes in 1911.



Critical or transition temperature (T_c):

The temperature at which the transition from normal state to superconducting state takes place on cooling in the absence of magnetic field is called **critical temperature or transition** temperature.

Table below gives the transition temperature of some superconducting materials

Element	T_c in K	Compound	T_c in K
Al	1.1	Pb ₂ Au	7
In	3.4	MoN	12
Ta	4.4	NbN	15
V	5.1	V ₃ Si	17

Properties of Superconductors:

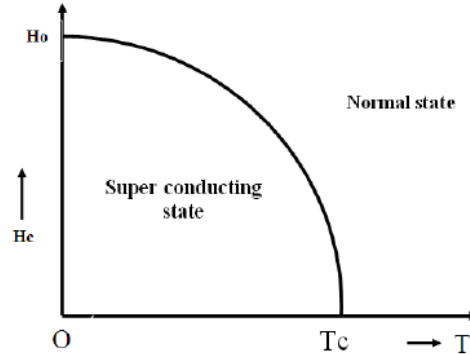
1. Critical magnetic field (H_c)

Superconductivity of a metal mainly depends on the temperature and strength of the magnetic field in which the metal is placed. Superconductivity disappears if the temperature of the specimen is raised above T_c or a strong enough magnetic field is applied. At temperatures below T_c , in the absence of magnetic field, the material is in superconducting state. When the strength of the magnetic field is increased to a critical value H_c the material loses its superconducting property. **Thus, we can define the critical magnetic field as, the strength of the applied magnetic field above which superconductivity in the material disappears.** The dependence of critical magnetic field upon the temperature is given by

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$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

The variation of H_c with respect to T is shown in figure.



2. The Meissner Effect

When a magnetic field smaller than critical field is applied to a superconducting specimen at a temperature below transition temperature T_c , the magnetic flux lines are expelled out of the superconductor. This phenomenon is called **Meissner effect**.

Consider a material in the normal state. When a magnetic field is applied to the material, the magnetic flux lines pass through the material. If the temperature (T) is reduced below the critical temperature (T_c) the magnetic flux lines will be expelled from inside the superconductor. The magnetic induction inside the material is given by,

$$B = \mu_0 (H + M)$$

where M is the magnetization in the material due to an applied magnetic field H .

At the temperature $T < T_c$, $B = 0$

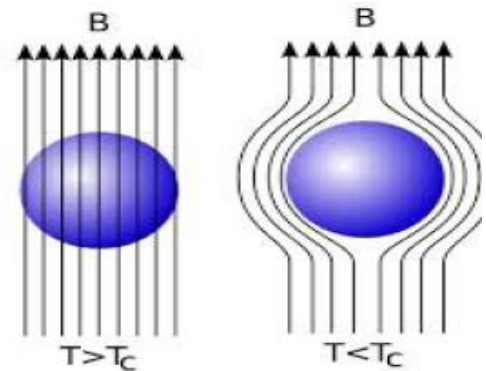
$$\Rightarrow \mu_0 (H + M) = 0 \Rightarrow H = -M \Rightarrow \frac{M}{H} = -1 \Rightarrow \chi = \frac{M}{H} = -1$$

where χ is the magnetic susceptibility and $\chi = -1$ indicates that the material in its superconducting state is a perfect diamagnetic material.

3. Effect of current

An electric current passing through the superconducting material itself produces a magnetic field, if the magnetic field produced exceeds critical magnetic field, it will destroy the superconductivity of the material. The maximum current that can be passed through the superconducting material without destroying its superconductivity is called critical current (I_c).

The expression for critical current is $I_c = 2\pi r H_c$ where r is radius of the superconducting wire and H_c is critical field. This equation is called **Silsbee's rule**. This puts a limit on the electric current that can be passed through a superconductor.



Types of superconductors: Type I and type II superconductors:

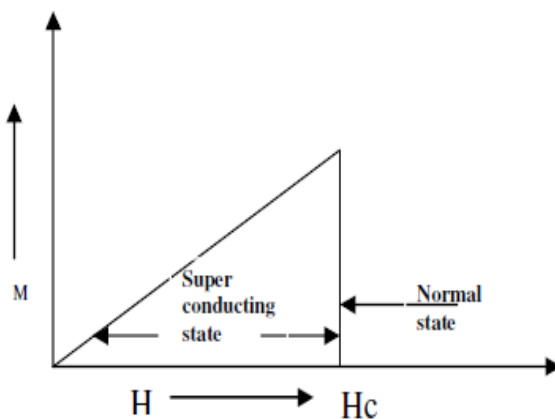
Based on the diamagnetic response superconductors can be classified into two types, they are

1. Type - I superconductors
2. Type - II superconductors

Type I superconductors:

Type-I superconductors also known as soft superconductors which follow a complete Meissner effect. Type I superconductors have a small critical magnetic field and the transition from the superconducting to the normal phase at the critical field is abrupt. When the applied magnetic field strength is gradually increased, at H_C the diamagnetism abruptly disappears and the transition from superconducting state to normal state takes place sharp at the critical field value as shown in figure. For all values of magnetic field above the critical field, the material shows finite resistivity, and the magnetic flux penetration is complete. In other words, Type I superconductors display Meissner effect completely. The low critical field value restricts these materials from its application in superconducting magnets.

Examples: Hg, Pb, Zn, and Sn etc..



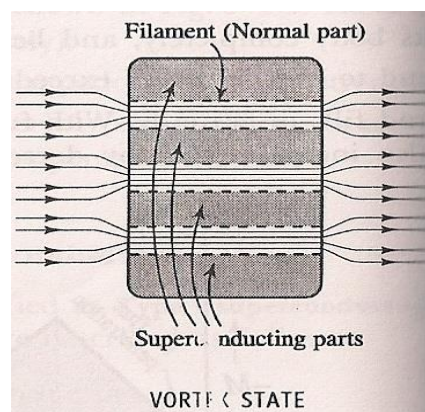
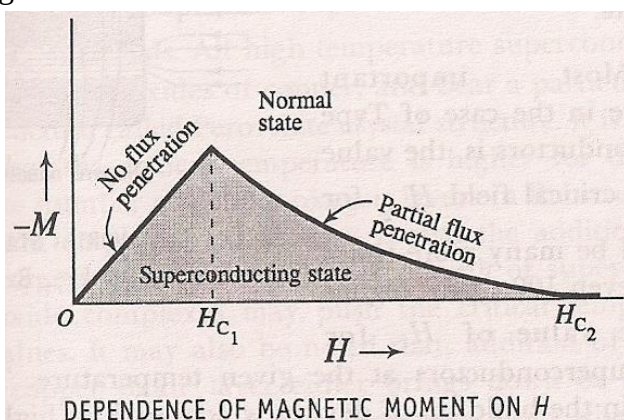
Type II Superconductors:

Type II superconductors, also known as hard superconductors, do not follow the complete Meissner effect. They have two critical fields lower critical field H_{C1} and upper critical field H_{C2} . The value of H_{C1} is small, H_{C2} is quite large. For the applied field less than H_{C1} , it expels the magnetic flux completely and behaves as a perfect diamagnetic. As applied field exceeds H_{C1} , the flux starts to penetrate through the superconductor. Between H_{C1} and H_{C2} the flux penetration increases with increase in field, but superconductors retain zero resistivity property. This state is called vortex state or mixed state. At H_{C2} the flux penetrates completely, and the specimen comes to normal state. The flux penetration takes place through thin channelized regions called filaments, the width of which increases with increase in field strength and at H_{C2} there will be a single filament.

Examples: - NbTi, Nb_3Sn

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Type-II superconductors are used in the manufacturing of the superconducting magnets of high magnetic fields above 10 Tesla.



Difference between type-1 and type 2 superconductors:

	Type-I	Type-II
1	Type-I superconductors are usually elements in their pure form	These are alloys or compounds
2	The transition to normal state occurs abruptly at a critical magnetic field (H_c)	The transition to normal state begins at H_{c1} and is completes only at H_{c2}
3	The value of critical magnetic field (H_c) is usually very small	Though the value of H_{c1} is small, H_{c2} is quite large.
4	They exhibit complete Meissner effect up to H_c	Meissner effect is complete up to H_{c1} when the magnetic flux starts penetrating into the superconductor. The flux penetration is completes only at H_{c2} when the material becomes normal conductor.
5	The superconductivity state is observed up to critical magnetic field (H_c)	The material remains in this resistance less state even in the intermediate state between H_{c1} and H_{c2}
6	Small value of H_c restricts their use in superconducting magnets.	Large value of H_{c2} makes it suitable for applications in superconducting magnets.

BCS theory

BCS theory of superconductor was put forward by Bardeen, Cooper and Schrieffer in 1957 and hence named as BCS theory. This theory could explain the phenomena of superconductivity.

The BCS theory is based upon the formation of Cooper pairs.

The main idea behind the BCS theory is the discovery of isotopic effect of superconductivity.

The isotope effect suggests that the current carrying electron in a superconductor do not move independently but instead they move interacting with lattice. (Electron lattice interaction via lattice deformation.)

The lattice is slightly deformed as an electron comes near the lattice and also lattice will displace from its position. The displacement of lattice is called **lattice distortion**.

The deformation produces increased positive charge at the lattice. Another electron moving through this positive region will be attracted by the greater concentration of the positive charge there.

Now, attraction is stronger than the repulsion between the electrons, so electrons are effectively coupled together into form a **Cooper pair** with the deformed lattice as the intermediary.

The electron – lattice – electron interaction does not keep the electrons a fixed distance. Electrons moving in opposite directions and their separation are as great as 10^{-6} m.

The binding energy of a Cooper pair called **energy gap E_g** is of the order of 10^{-3} eV.

$$E_g \text{ at } 0 \text{ K} = E_g(0) = 3.53kT_C$$

Where k – Boltzmann constant & T_C – Critical temperature.

At room temperature (300 K) energy is $kT = 0.026 \text{ eV}$ which is higher than the binding energy of cooper pairs which is sufficient to break the cooper pair. This is the reason why superconductor phenomena is observed only at low temperature.

The electrons in a cooper pair have opposite spins, so the pair has a total spin of zero. As a result, the electron pairs in a superconductor are **bosons**.

Linear momentum of the electrons in a cooper pair is equal and opposite and total momentum is zero.

Since electrons are bosons all electrons are exist in same energy state or quantum state. All these electrons involve in current in a superconductor. To alter this current requires a large energy, the current persists indefinitely.

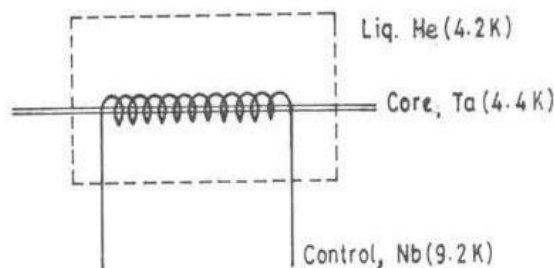
Applications of superconductors:

The major applications that are based on the properties of the superconducting phase are

1. Superconducting memories and switches (Cryotron)
2. Magnetic levitation
3. Lossless power transmission
4. Superconducting magnets
5. Superconducting quantum interference devices (SQUIDS)

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1. **Superconducting switch - Cryotron:** The disappearance of superconductivity for magnetic fields higher than the critical field is the principle used in the construction of a **cryotron**.



It consists of a core wire A around which a control coil B is wound. The core A is made of tantalum ($T_c=4.4$ K) and control coil B is made of niobium ($T_c=9.2$ K) or lead ($T_c=7.2$ K). The whole assembly is maintained at liquid helium temperature (4.2 K). At this temperature, both the control coil and the core wire are in the superconducting phase. Hence, the resistance of the core wire will be zero. A current can be passed through the control coil to produce a magnetic field sufficient to make the core wire 'normal'. Thus, the core wire can be made to possess zero or finite resistance depending on the control current being 'off' or 'on' respectively. These two states of the core wire may be considered as the ON and OFF states. Thus, cryotrons can be used as switches.

2. **Magnetic levitation:** The principle of the repulsion of magnetic flux from a superconductor can be used in **magnetic levitation** applications. When a magnet is brought near a superconductor, there will be a repulsion and the superconductor tries to move away from the magnet. Because of this superconducting wheel experience reduced friction on magnetic tracks.
3. **Lossless power transmission:** The transmission of electrical power on a large scale is limited due to the restrictions on the current carrying capacity of transmission lines and their maintenance. Loss of power due to power dissipation in transmission lines due to the finite resistance offered by the cables is quite considerable. If the transmission lines are made of materials in their superconducting state, their current carrying capacity increases. Since these lines do not offer any resistance to the flow of electric current, there is no power dissipation due to joule heating. Thus, power losses are minimized and current carrying capacity is improved. This is the principle used in **lossless power transmission**.
4. **Superconducting magnets:** Use of superconducting coils in the electromagnets can enhance the magnetic field generated as they can carry large currents with practically no dissipation. This is the principle used in the construction of **superconducting magnets**.

High Temperature Superconductors (HTS):

High-temperature superconductors (HTSs) are materials that exhibit superconductivity at temperatures significantly higher than conventional superconductors, typically above 77 K (the boiling point of liquid nitrogen).

Discovery: The first HTS material was a copper-oxide ceramic discovered by Georg Bednorz and K. Alex Müller, for which they were awarded the Nobel Prize in Physics in 1987. This breakthrough sparked a race to find materials with even higher critical temperatures.

In a couple of years, spectacular results were reported with T_c up to 125 K. This led to the hope that it might be possible to develop materials that are superconductors at room temperature. Discovery of such materials would mean large scale application of the phenomenon.

All high temperature superconductors are different types of oxides of copper and bear a particular type of crystal structure called Perovskite crystal structure. It is interesting that the critical temperature is higher for those which have a greater number of copper-oxygen layers in the structure unit. So far high temperature superconductivity is observed only by nonmetals like ceramics. The major technical challenge is therefore to get them in proper shape such as wires and ribbons which facilitates the flow of electricity over a long range.

Some of these superconducting materials are:

$\text{YBa}_2\text{Cu}_3\text{O}_7$ ($T_c = 90 \text{ K}$),

$\text{Bi}_2\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$ ($T_c = 110 \text{ K}$)

$\text{TlBa}_2\text{Ca}_3\text{Cu}_4\text{O}_{11}$ ($T_c = 120 \text{ K}$)

$\text{Tl}_2\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$ ($T_c = 125 \text{ K}$)

The discovery of high temperature superconductivity is an important achievement as it eliminates the difficulty of maintaining low temperatures using liquid helium. Liquid nitrogen is a much cheaper and more accessible coolant than the liquid helium needed for conventional superconductors, making practical applications more feasible

Applications: High-temperature superconductor applications include powerful magnets for MRI machines and fusion reactors, efficient electric motors and generators, energy-saving power transmission lines and fault current limiters, and next-generation transportation via magnetic levitation (Maglev) systems. They also have potential in energy storage, advanced microwave devices, and research tools like particle accelerators and high-field magnets for spectroscopy.

Numerical Problems

1. Calculate the drift velocity and thermal velocity of conduction electrons in copper at a temperature of 300K, when a copper wire of length 2 m and resistance $0.02\ \Omega$ carries a current of 15A. Given the mobility of free electrons in copper is $4.3 \times 10^{-3}\ \text{m}^2/\text{Vs}$.
2. A uniform silver wire has a resistivity of $1.54 \times 10^{-8}\ \Omega\text{m}$ at room temperature. For an electric field along the wire of 100 V/m, compute the drift velocity of an electron and the mobility assuming that there is 5.8×10^{28} conduction electrons/ m^3 .
3. The relaxation time of conduction electrons in copper is $2.6 \times 10^{-14}\text{s}$. If copper has a resistivity of $1.6 \times 10^{-8}\ \text{Ohm m}$, calculate the number of free electrons per unit volume and Fermi velocity. Assume Fermi energy as 7 eV.
4. Estimate the relaxation time of conduction electrons in silver from the following data: Resistivity = $1.6 \times 10^{-8}\ \text{ohm m}$ Density = $10.5 \times 10^3\ \text{kg/m}^3$. Atomic weight = 107.88. Avogadro number = 6.023×10^{23} per gm mole. (6.023×10^{26} per kg mole).
5. A uniform wire has a resistivity of $2 \times 10^{-8}\ \Omega\text{m}$ at room temperature. For an electric field of 200 V/m along the wire compute the average drift velocity of the electrons assuming a carrier concentration of $6 \times 10^{28}\ \text{m}^{-3}$. Also calculate the mobility and relaxation time of electrons.
6. The resistivity of copper at 0°C is $1.6 \times 10^{-8}\ \Omega\ \text{m}$ and changes to $2.2 \times 10^{-8}\ \text{ohm m}$ at 100°C . By what factor does the relaxation time of conduction electron change?
7. Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K.
8. Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above Fermi energy is occupied.
9. The critical temperature and critical magnetic field at 0 K for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss?
10. Superconducting tin has a critical magnetic field of 217 gauss at 2 K. If the critical temperature for superconducting transition for tin is 3.7 K, find the critical magnetic field at 3 K.
11. A sample of silicon semiconductor is doped with 10^{22} phosphorous atoms. Calculate its conductivity if mobility of electrons is $0.07\ \text{m}^2/\text{V/s}$. What is the Hall voltage if this semiconductor with a thickness of $100\ \mu\text{m}$ and carrying a current of 1 mA is placed perpendicular to a magnetic field of 0.1T.?

Question Bank: UNIT III: Conductors and Superconductors

12. What are the basic assumptions of the classical free electron theory?
13. Obtain an expression for the electrical conductivity of a metal based on the classical free electron theory.
14. What are the drawbacks (failure) of classical free electron theory?
15. Mention the important differences between classical free electron theory and quantum free electron theory.
16. Explain the term drift velocity in metals.
17. Distinguish between relaxation time and mean collision time.
18. Mention the various reasons for the scattering of electrons in a metal and their contribution to the electrical resistivity of metals.
19. Define the terms Fermi factor and Fermi energy.
20. Explain the Fermi Dirac distribution of electrons with respect to different temperature conditions.
21. What is a Fermi factor? Explain the Fermi factor for different temperature conditions.
22. What are superconductors? Explain its characteristic properties.
23. Explain Critical field and Meissner effect in superconductors.
24. Explain the classifications of superconductors.
25. Distinguish between Type-I and Type- II superconductors.
26. Explain Type-I and Type-II superconductors.
27. Write a note on BCS theory.
28. Write a note on various applications of superconductors.
29. Write a note on high temperature superconductors.

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